INTERNATIONAL SEARCH REPORT

International application No.
PCT/EP2022/063227

A. CLASSIFICATION OF SUBJECT MATTER

G01S 17/89(2020.01)i; ***G01S 17/42***(2006.01)i; ***G01S 17/04***(2020.01)i; ***G01S 7/48***(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, INSPEC, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	DE 102020004791 A1 (DAIMLER AG [DE]) 26 November 2020 (2020-11-26) paragraphs [0006], [0014] - [0017], [0011]	1-10
A	DE 102016224764 A1 (BOSCH GMBH ROBERT [DE]) 14 June 2018 (2018-06-14) paragraph [0034]; figure 4	2
A	US 2019361098 A1 (HOLLMANN JOSEPH [US] ET AL) 28 November 2019 (2019-11-28) paragraph [0019]	5
A	US 2020056973 A1 (KNOX RONALD [AU] ET AL) 20 February 2020 (2020-02-20) paragraph [0257]	6

☐ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
13 September 2022	**20 September 2022**
Name and mailing address of the ISA/EP **European Patent Office** **p.b. 5818, Patentlaan 2, 2280 HV Rijswijk** **Netherlands** Telephone No. **(+31-70)340-2040** Facsimile No. **(+31-70)340-3016**	Authorized officer **Ferrara, Michele** Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/EP2022/063227

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
DE 102020004791 A1	26 November 2020	NONE	
DE 102016224764 A1	14 June 2018	NONE	
US 2019361098 A1	28 November 2019	US 2019361098 A1	28 November 2019
		WO 2019226487 A1	28 November 2019
US 2020056973 A1	20 February 2020	AU 2008323626 A1	22 May 2009
		CA 2705830 A1	22 May 2009
		CA 2993208 A1	22 May 2009
		CA 3103407 A1	22 May 2009
		CN 101952709 A	19 January 2011
		CN 105185026 A	23 December 2015
		CN 109255924 A	22 January 2019
		EP 2217911 A1	18 August 2010
		EP 3082117 A1	19 October 2016
		EP 3367358 A2	29 August 2018
		HK 1217807 A1	20 January 2017
		JP 5662154 B2	28 January 2015
		JP 6030092 B2	24 November 2016
		JP 6523220 B2	29 May 2019
		JP 2011503581 A	27 January 2011
		JP 2015014606 A	22 January 2015
		JP 2016218075 A	22 December 2016
		KR 20100110783 A	13 October 2010
		KR 20160067187 A	13 June 2016
		KR 20170018086 A	15 February 2017
		TW 200925577 A	16 June 2009
		TW 201538953 A	16 October 2015
		TW 201734428 A	01 October 2017
		US 2011058167 A1	10 March 2011
		US 2015241330 A1	27 August 2015
		US 2017276588 A1	28 September 2017
		US 2020056973 A1	20 February 2020
		WO 2009062256 A1	22 May 2009